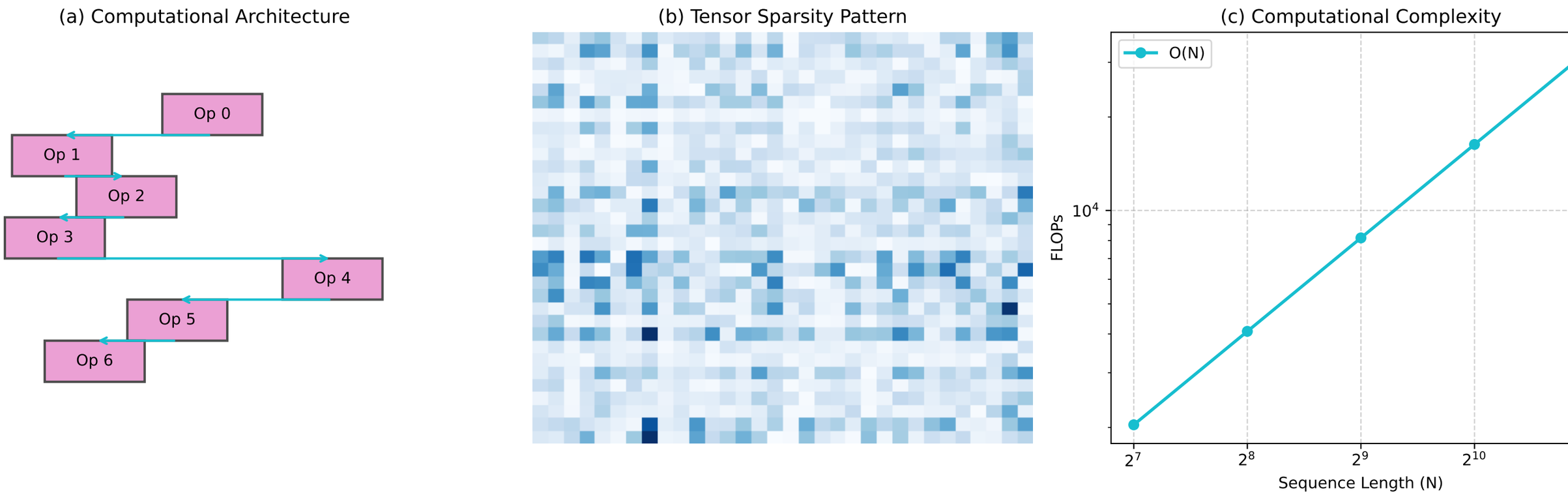Module Architecture: engine